

Checkpoint 1: When Is a Positive Really Positive?

! Important

Due 10/18 (12 points). See [Project Central](#) for the full project overview. Make sure you have addressed any feedback from Checkpoint 0.

Goal

Use conditional probability and Bayes' rule to answer: *If someone served by my public health program gets a positive result, how likely is it that they truly have the condition?*

Course topics used: events, conditional probability, the Law of Total Probability, Bayes' rule, independence.

Your scenario

Use the input values from your [Checkpoint 0](#) table, with any corrections from my feedback. Every value should come from your source packet or from a course value on your scenario card.

Scenario A

Community breast cancer screening program

- **Events:** D = has breast cancer; T^+ = the mammogram is read as abnormal.
- **Inputs:** prevalence (course value), sensitivity, and specificity from Lee et al. (2023).
- **Program question:** What does an abnormal read mean for a woman screened on the van?

Scenario B

County COVID-19 testing and contact tracing

- **Events:** D = infected with SARS-CoV-2; T^+ = the rapid antigen test is positive.
- **Inputs:** prevalence (course value), and sensitivity and specificity for people without symptoms from the Cochrane review.
- **Program question:** How much should a resident without symptoms trust a positive rapid test?

Scenario C

State newborn heart screening program

- **Events:** D = has a critical congenital heart defect (CCHD); T^+ = the pulse oximetry screen is positive.
- **Inputs:** prevalence, sensitivity, and false-positive rate from the Cochrane review. Your false-positive rate is already $P(T^+ | D^c)$, and your specificity is $1 - (\text{false-positive rate})$.
- **Program question:** What does a positive screen mean for a newborn's family?

Tasks

1. Set up the problem

- Copy your events and your three Checkpoint 1 probability statements from Checkpoint 0, with any corrections from my feedback. Keep the source for each value.
- Add the one conditional probability you did not record in Checkpoint 0, using the complement rule:
 - **Scenarios A and B:** the false-positive rate, $P(T^+ | D^c) = 1 - P(T^- | D^c)$.
 - **Scenario C:** the specificity, $P(T^- | D^c) = 1 - P(T^+ | D^c)$.
- Draw a tree diagram **or** a 2×2 table for a hypothetical population of 100,000 people screened by your program. Round each count to a whole number of people.

2. Solve it analytically

- Use the Law of Total Probability to find $P(T^+)$.
- Use Bayes' rule to find the **positive predictive value** $P(D | T^+)$ and the **negative predictive value** $P(D^c | T^-)$.
- Show your work. You may use R as a calculator, but write out each formula first.

💡 Check yourself (Scenario A only, not graded)

Lee et al. (2023) report an abnormal interpretation rate of 8.3% and a PPV1 of 6.9% for DBT. Your $P(T^+)$ and $P(D | T^+)$ should be close to these values. If they are, Bayes' rule just reproduced numbers from a published study!

3. Simulate it

- Simulate a population of at least 100,000 people in R. Assign each person a disease status, then a test result that depends on their disease status.
- Estimate PPV and NPV from your simulated population and compare them with your analytic answers.
- In 1–2 sentences, explain why the simulated and analytic values are close but not identical.

Starter code (replace the placeholder input values with your own). This follows the mammogram example from Lesson 4.

```
set.seed(550) ①
reps <- 100000 ②
p_D <- 0.01 ③
p_pos_if_D <- 0.90
p_pos_if_no_D <- 0.05

people <- tibble(person = 1:reps) |> ④
  mutate(D = sample(c(TRUE, FALSE), n(), replace = TRUE, ⑤
    prob = c(p_D, 1 - p_D)),
    pos_if_D = sample(c(TRUE, FALSE), n(), replace = TRUE, ⑥
    prob = c(p_pos_if_D, 1 - p_pos_if_D)),
    pos_if_no_D = sample(c(TRUE, FALSE), n(), replace = TRUE, ⑦
    prob = c(p_pos_if_no_D, 1 - p_pos_if_no_D)),
    positive = if_else(D, pos_if_D, pos_if_no_D)) ⑧

people |> ⑨
  filter(positive) |>
  summarise(ppv_sim = mean(D))

people |> ⑩
  filter(!positive) |>
  summarise(npv_sim = mean(!D))
```

- ① Set a seed so you get the same simulated population every time you render.

- ② The number of simulated people.
- ③ Replace these with your values: $P(D)$, $P(T^+ | D)$, and $P(T^+ | D^c)$. For Scenarios A and B, $P(T^+ | D^c)$ is 1 minus your specificity. For Scenario C, it is your false-positive rate.
- ④ One row per simulated person.
- ⑤ Does this person have the condition? `TRUE` has chance `p_D`.
- ⑥ What the test **would** show if this person has the condition: positive with chance `p_pos_if_D`.
- ⑦ What the test **would** show if this person does not have the condition: a false positive with chance `p_pos_if_no_D`.
- ⑧ If the person has the condition, keep `pos_if_D`. Otherwise keep `pos_if_no_D`. This is how the test result depends on disease status.
- ⑨ Keep only the people with a positive test, then find the proportion who have the condition. Because `TRUE` counts as 1 and `FALSE` as 0, `mean(D)` is a proportion.
- ⑩ Keep only the people with a negative test (`!` means “not”), then find the proportion who do **not** have the condition.

💡 Rare conditions need big populations

If your prevalence is very small (Scenario C has about 6 cases per 10,000 babies), a population of 100,000 has only about 60 true cases, so your simulated PPV will bounce around. Try `reps <- 1000000` and see whether your simulated values get closer to your analytic answers.

4. Explore: how does prevalence matter?

- Compute the analytic PPV for prevalence values from 0.01% to 50%, keeping sensitivity and specificity fixed.
- Plot PPV (y-axis) against prevalence (x-axis), and mark your scenario’s prevalence on the plot.
- In 2–3 sentences, describe what happens to PPV as prevalence changes. Then describe a group or setting your program might serve where the prevalence would be higher or lower than your scenario’s value (for example, women with a family history of breast cancer, residents tested during a COVID-19 surge, or babies whose prenatal ultrasound showed a possible heart problem). How would a positive result mean something different there?

To make the prevalence values, you can use `seq()`, which makes an evenly spaced sequence of numbers:

```
prev_values <- seq(from = 0.0001, to = 0.5, by = 0.0001)
```

This gives 5,000 values from 0.0001 (0.01%) to 0.5 (50%), in steps of 0.0001. Put them in a tibble, use `mutate()` to compute the PPV for each one, and plot with `ggplot()`.

💡 Tip: seeing small prevalences

All three scenarios have a prevalence below 1%, so on a regular x-axis your point sits squished against the left edge. Adding `+ scale_x_log10()` to your plot puts the x-axis on a log scale, which spreads out the small values (each step to the right multiplies the prevalence by 10).

To mark your scenario's point, you can add `+ annotate("point", x = your_prevalence, y = your_ppv)`, which adds a single point at the values you give it.

5. Say it

Your program sends a letter or text message to every person with a positive result. Write a short paragraph (4–6 sentences) for that message. Explain what the result means using natural frequencies (e.g., “Out of 1,000 people who get a positive result, about ____ actually have...”) rather than probability notation, and say what the next step is (a follow-up appointment, a confirmatory test, or a heart ultrasound).

Then answer your **Checkpoint 1 guiding question** from Checkpoint 0 in one sentence.

Submission

Submit a rendered HTML file (use `code-fold: true`) and your `.qmd`. Make sure the `qmd` and `html` are named `lastname_firstinitial_checkpoint_01.qmd` and `lastname_firstinitial_checkpoint_01.html` or `lastname1_lastname2_checkpoint_01.qmd` and `lastname1_lastname2_checkpoint_01.html`. One submission per team. Either person in the pair can submit it.

You can download the [.qmd editing file for this checkpoint here](#). The link takes you to GitHub, where you can click the download button (the arrow pointing down) near the top right of the file.

The editing file has the headings and tasks, but not the detailed instructions on this page. **Please do not remove anything from the editing file.** You will only add your work to it.

Rubric (12 points)

The full rubric, with every item and its points, is in the [Checkpoint 1 rubric spreadsheet](#).

Section	Points
Set up the problem (events, probability statements, tree or 2×2 table)	2.5
Solve it analytically ($P(T^+)$, PPV, NPV with formulas)	3
Simulate it (code, comparison with analytic answers, explanation)	2.5
Explore: how does prevalence matter? (plot and discussion)	2
Say it (program message and guiding question)	1.5
General / document quality	0.5